

Preparation Plan – Tech 3 (Master’s Level)

Instituto Itaú de Ciência, Tecnologia e Inovação

Inova Talentos

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Comprehensive Technical Structure

Block 1 — Core Foundations

Quantum Computing

- Quantum optimization (QAOA, VQE)
- Parameterized quantum circuits
- Simulation of quantum systems

Mathematical Foundations

- Linear Algebra (vector spaces, operators, eigenvalues)
- Quantum Mechanics (postulates, Hermitian operators, measurement theory)

Block 2 — Scientific Programming

- Advanced Python
- Scientific libraries:
 - NumPy
 - SciPy
 - Matplotlib
- Object-Oriented Programming (OOP)

Block 3 — Quantum Frameworks

- Qiskit
- PennyLane
- Cirq

Applications

- Quantum circuit simulation
- Execution on quantum hardware
- Integration with classical machine learning models

Block 4 — Machine Learning and Generative AI

Frameworks

- PyTorch
- TensorFlow
- JAX

Models

- Transformers (LLMs)
- Variational Autoencoders (VAE)
- Diffusion Models

Techniques

- Fine-tuning
- Model evaluation

Block 5 — Optimization Methods

Classical Optimization

- Gradient Descent
- Adam
- COBYLA
- SPSA

Note: Classical optimization plays a central role in hybrid quantum-classical algorithms, where parameters of quantum circuits are iteratively adjusted.

Block 6 — Quantum Machine Learning (QML)

- Quantum Neural Networks (QNNs)
- Variational Quantum Eigensolver (VQE)
- Quantum Approximate Optimization Algorithm (QAOA)

Integrations

- Qiskit Machine Learning
- PennyLane AI

Block 7 — Advanced and Desirable Skills

Generative AI

- Reinforcement Learning from Human Feedback (RLHF)
- Prompt engineering and dataset design

Infrastructure

- GPU computing
- Experimental pipelines

MLOps

- MLflow
- Weights & Biases
- Data Version Control (DVC)

Block 8 — Scientific Production and Reproducibility

- Jupyter Notebooks
- LaTeX
- Git and GitHub
- Reproducible research practices

Learning Strategy (Realistic Roadmap)

Stage 1 — Foundations

- Scientific Python and linear algebra
- Fundamental quantum mechanics

Stage 2 — Intermediate Development

- Quantum circuits using Qiskit or PennyLane
- Numerical optimization methods

Stage 3 — Advanced Topics

- Variational algorithms (VQE, QAOA)
- Integration with machine learning frameworks (PyTorch, JAX)

Stage 4 — Specialization

- Quantum Machine Learning (QML)
- Generative AI and end-to-end pipeline development

Chapter 1

Core Foundations

This block focuses on the fundamental structure of modern quantum computing:

How to represent, evolve, and optimize quantum systems using linear algebra and parametrized quantum circuits

1.1 Quantum Computing

Quantum computing is a computational paradigm that leverages the principles of quantum mechanics to process information. Unlike classical computers, which use bits as the basic unit of information, quantum computers use quantum bits (qubits), enabling fundamentally new forms of computation.

1.1.1 Qubit and Hilbert Space

A qubit is not just a binary unit, but a vector in a complex Hilbert space, where its state is described by a superposition of basis states, enabling the rich structure of quantum computation.

A qubit represented by a vector in a 2-dimensional complex vector space:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (1.1)$$

with normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1 \quad (1.2)$$

This naturally introduces:

- Linear algebra
- Quantum probability

1.1.2 Operators and Quantum Gates

In quantum computing, operators and quantum gates are fundamentally the same mathematical objects viewed from different perspectives.

An operator is a linear transformation acting on vectors in a Hilbert space. In the context of a qubit. The evolution of a quantum state is given by:

$$|\psi'\rangle = \hat{U}|\psi\rangle \quad (1.3)$$

where:

- $\hat{U}$ is a unitary operator
- $\hat{U}^\dagger \hat{U} = I$

Examples of quantum gates:

- Pauli-X
- Hadamard
- CNOT

They operate on qubits and are analogous to classical logic gates, but they obey the principles of quantum mechanics.

1.1.3 Single-Qubit Gates

Single-qubit gates are the fundamental building blocks of quantum circuits. They act on a single qubit and correspond to unitary transformations in a 2-dimensional complex Hilbert space.

1.1.3.1 Pauli-X Gate (NOT Gate)

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (1.4)$$

What does it do? It **flips the state of a qubit**, just like a classical NOT gate:

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle \quad (1.5)$$

Why is it called a NOT gate?

In classical computing:

$$\text{NOT}(0) = 1$$

$$\text{NOT}(1) = 0$$

The Pauli-X gate performs exactly the same operation, but on a quantum bit (qubit).

1.1.3.2 Action on a General State

If a qubit is in a superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

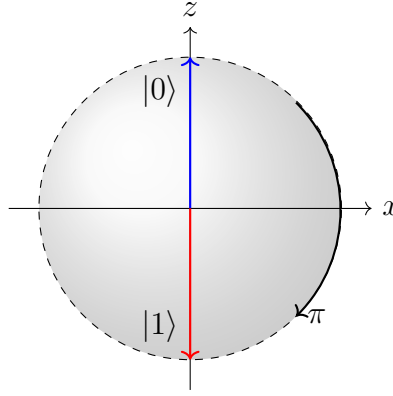
Applying the Pauli-X gate:

$$X|\psi\rangle = \alpha|1\rangle + \beta|0\rangle$$

It swaps the amplitudes of $|0\rangle$ and $|1\rangle$.

1.1.3.3 Geometric Interpretation (Bloch Sphere)

- The Pauli-X gate corresponds to a rotation of π radians around the x -axis.
 - It moves the qubit from the north pole to the south pole of the Bloch sphere.
-



Pauli-X: rotation around x -axis

Figure 1.1: Bloch sphere representation of the Pauli-X gate

1.1.3.4 Pauli-Y Gate

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad (1.6)$$

$$Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle \quad (1.7)$$

Example: Single-Qubit Parametrized Quantum Circuit (PQC)

A simple parametrized quantum circuit is defined as:

$$|\psi(\theta)\rangle = R_y(\theta) |0\rangle$$

The rotation operator around the y -axis is given by:

$$R_y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$

Applying $R_y(\theta)$ to the initial state $|0\rangle$:

$$|\psi(\theta)\rangle = \cos(\theta/2) |0\rangle + \sin(\theta/2) |1\rangle$$

Measurement Probabilities

If we measure in the computational basis $\{|0\rangle, |1\rangle\}$:

$$P(0) = \cos^2(\theta/2)$$

$$P(1) = \sin^2(\theta/2)$$

Thus, the parameter θ directly controls the measurement probabilities.

Python Implementation (Qiskit)

```
import numpy as np
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator

theta = np.pi / 3

qc = QuantumCircuit(1, 1)
qc.ry(theta, 0)
qc.measure(0, 0)

simulator = AerSimulator()
result = simulator.run(qc, shots=10000).result()
counts = result.get_counts()

print(counts)

p1 = np.sin(theta/2)**2
p0 = np.cos(theta/2)**2

print("P(1):", p1)
print("P(0):", p0)
```

1.1.3.5 Output from Algorithm: Qiskit

```
python main.py
Measurement counts: {'1': 2482, '0': 7518}
```

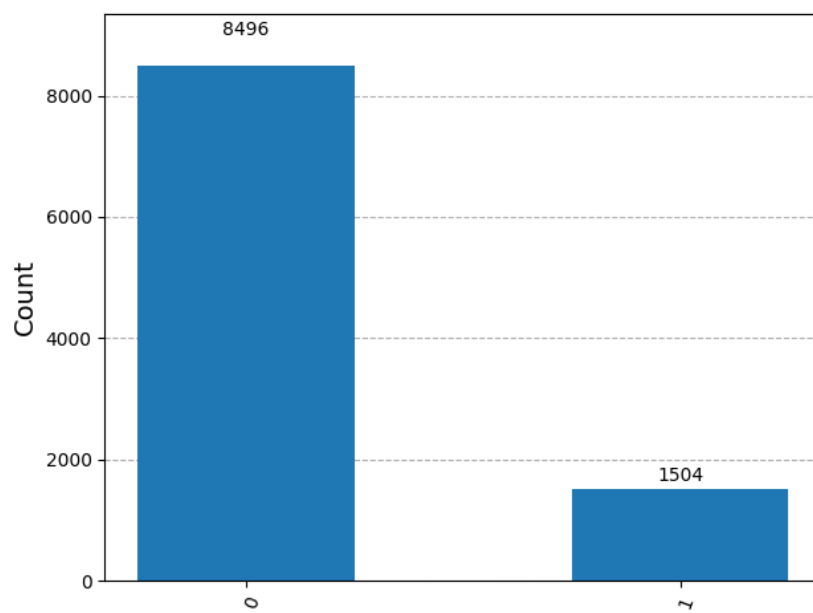


Figure 1.2: Example to Qiskit

Theoretical $P(1)$: 0.2500

Theoretical $P(0)$: 0.7500

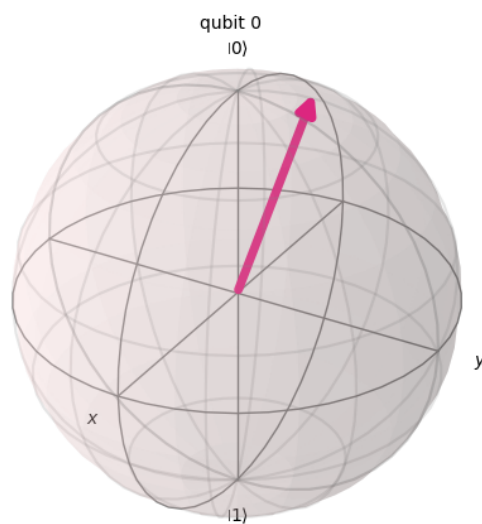


Figure 1.3: Qiskit Bloch Sphere.

1.1.3.6 Pauli-Z Gate

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (1.8)$$

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle \quad (1.9)$$

1.1.3.7 Hadamard Gate

The Hadamard gate creates a superposition of basis states.

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (1.10)$$

Its action on the computational basis is given by:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (1.11)$$

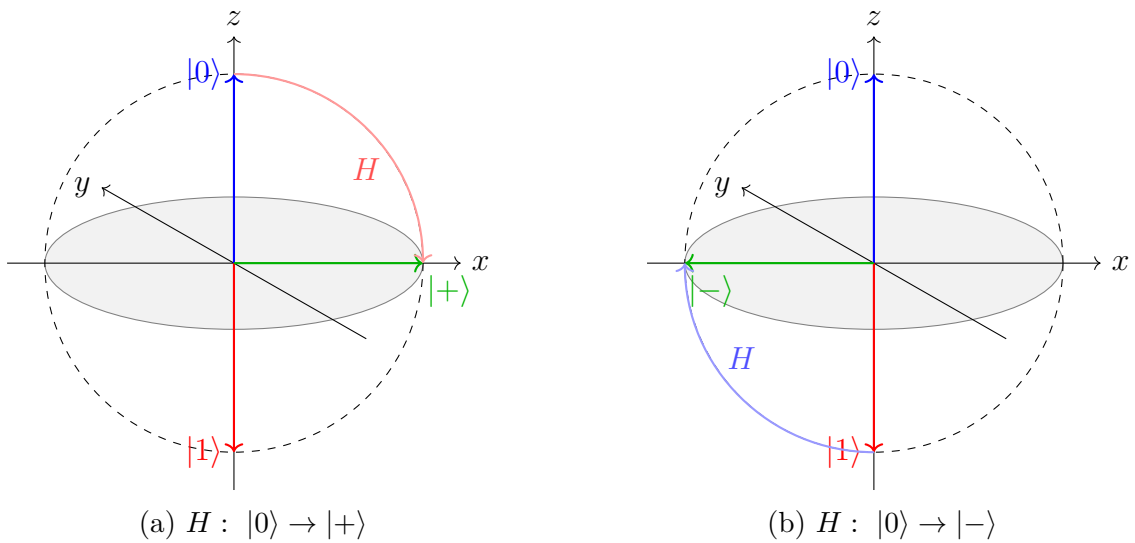


Figure 1.4: Bloch sphere representation of the Hadamard transformation

Bloch Sphere Representation of the State $|+\rangle$

Algebraic Definition

The state $|+\rangle$ is defined as:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad (1.12)$$

This represents a superposition with:

- Equal probability amplitudes for $|0\rangle$ and $|1\rangle$
- No relative phase between the components

Measurement in the computational basis yields:

$$P(0) = \frac{1}{2}, \quad P(1) = \frac{1}{2} \quad (1.13)$$

General Qubit State on the Bloch Sphere

Any single-qubit pure state can be written as:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle \quad (1.14)$$

where:

- $\theta \in [0, \pi]$ is the polar angle
- $\phi \in [0, 2\pi)$ is the azimuthal angle

Mapping $|+\rangle$ onto the Bloch Sphere

Comparing:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad (1.15)$$

we identify:

$$\cos\left(\frac{\theta}{2}\right) = \frac{1}{\sqrt{2}} \quad (1.16)$$

$$\sin\left(\frac{\theta}{2}\right) = \frac{1}{\sqrt{2}} \quad (1.17)$$

which implies:

$$\theta = \frac{\pi}{2} \quad (1.18)$$

Since there is no complex phase:

$$e^{i\phi} = 1 \quad \Rightarrow \quad \phi = 0 \quad (1.19)$$

Geometric Interpretation

Thus, the state $|+\rangle$ corresponds to:

- $\theta = \frac{\pi}{2}$ (equatorial plane)
- $\phi = 0$ (aligned with the x -axis)

$$|+\rangle \longleftrightarrow \text{+x direction on the Bloch sphere} \quad (1.20)$$

Interpretation of Superposition

The computational basis states correspond to:

$$|0\rangle \leftrightarrow \text{north pole } (z = +1) \quad (1.21)$$

$$|1\rangle \leftrightarrow \text{south pole } (z = -1) \quad (1.22)$$

A superposition does not lie “between” these states linearly. Instead:

- Amplitudes determine the polar angle θ
- Relative phase determines the azimuthal angle ϕ

Role of Relative Phase

Consider:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad (\phi = 0) \quad (1.23)$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (\phi = \pi) \quad (1.24)$$

Both states yield equal measurement probabilities, but correspond to:

$$|+\rangle \leftrightarrow +x \text{ direction} \quad (1.25)$$

$$|-\rangle \leftrightarrow -x \text{ direction} \quad (1.26)$$

Thus, the relative phase rotates the state around the Bloch sphere.

Therefore:

$|+\rangle$ is not simply “between” $|0\rangle$ and $|1\rangle$, but a state whose interference structure aligns with the x -axis.

1.1.4 Phase Gates (S and T)

In quantum computing, phase gates are single-qubit gates that modify the *relative phase* between basis states without changing measurement probabilities.

1.1.4.1 S Gate (Phase Gate)

The S gate applies a phase of $\frac{\pi}{2}$ to the $|1\rangle$ component:

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad (1.27)$$

Action on basis states:

$$S |0\rangle = |0\rangle \quad (1.28)$$

$$S |1\rangle = i |1\rangle \quad (1.29)$$

Thus, the S gate introduces a relative phase shift of $\frac{\pi}{2}$.

1.1.4.2 T Gate ($\pi/8$ Gate)

The T gate applies a phase of $\frac{\pi}{4}$:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \quad (1.30)$$

Action on basis states:

$$T |0\rangle = |0\rangle \quad (1.31)$$

$$T |1\rangle = e^{i\pi/4} |1\rangle \quad (1.32)$$

1.1.4.3 General Phase Gate

Both S and T are special cases of the general phase gate:

$$P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \quad (1.33)$$

$$S = P\left(\frac{\pi}{2}\right), \quad (1.34)$$

$$T = P\left(\frac{\pi}{4}\right) \quad (1.35)$$

1.1.4.4 Effect on a Superposition State

Consider the state:

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle \quad (1.36)$$

Applying the S gate:

$$S |\psi\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}} \quad (1.37)$$

Applying the T gate:

$$T |\psi\rangle = \frac{|0\rangle + e^{i\pi/4}|1\rangle}{\sqrt{2}} \quad (1.38)$$

1.1.4.5 Bloch Sphere Interpretation

Phase gates correspond to rotations around the Z-axis:

$$S = R_z\left(\frac{\pi}{2}\right), \quad (1.39)$$

$$T = R_z\left(\frac{\pi}{4}\right) \quad (1.40)$$

These rotations change the phase (longitude) without affecting measurement probabilities.

1.1.4.6 Key Properties

$$S^2 = Z \quad (1.41)$$

$$T^2 = S \quad (1.42)$$

$$T^4 = Z \quad (1.43)$$

$$T^8 = I \quad (1.44)$$

1.1.4.7 Summary

- Phase gates modify relative phase, not amplitudes
- S gate applies a $\frac{\pi}{2}$ phase shift
- T gate applies a $\frac{\pi}{4}$ phase shift
- Both correspond to rotations around the Z-axis
- The T gate is essential for universal quantum computation

1.1.5 Multi-Qubit Gates

Multi-qubit gates are fundamental in quantum computing because they allow interactions between qubits, enabling the creation of entanglement, a uniquely quantum resource.

1.1.5.1 General Idea

A single-qubit state is given by:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1.45)$$

A two-qubit state lives in a larger Hilbert space:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle \quad (1.46)$$

The dimension of the Hilbert space grows exponentially:

- 1 qubit $\rightarrow$ dimension 2

- 2 qubits $\rightarrow$ dimension 4
- n qubits $\rightarrow$ dimension 2^n

1.1.5.2 Controlled-NOT (CNOT) Gate

The CNOT gate is the most important two-qubit gate.

Definition

It consists of:

- One control qubit
- One target qubit

Its action is:

$$|c, t\rangle \rightarrow |c, t \oplus c\rangle \quad (1.47)$$

Action on Basis States

$$|00\rangle \rightarrow |00\rangle \quad (1.48)$$

$$|01\rangle \rightarrow |01\rangle \quad (1.49)$$

$$|10\rangle \rightarrow |11\rangle \quad (1.50)$$

$$|11\rangle \rightarrow |10\rangle \quad (1.51)$$

1.1.5.3 Controlled-NOT (CNOT) Gate

The CNOT gate flips the target qubit if the control qubit is in state $|1\rangle$.

$$\text{CNOT}|10\rangle = |11\rangle \quad (1.52)$$

Matrix representation:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (1.53)$$

1.1.5.4 SWAP Gate

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1.54)$$

$$\text{SWAP}|ab\rangle = |ba\rangle \quad (1.55)$$

1.1.5.5 Toffoli Gate (CCNOT)

The Toffoli gate has two control qubits and one target qubit. The target qubit is flipped only if both control qubits are in state $|1\rangle$.

1.1.5.6 Properties of Quantum Gates

- All quantum gates are unitary (reversible transformations).
- They preserve normalization of the quantum state.
- They enable superposition and entanglement.

1.1.5.7 Applications

Quantum gates are essential for:

- Quantum algorithms (e.g., Shor's algorithm, Grover's algorithm)
- Quantum cryptography
- Quantum simulation

1.1.5.8 Quantum Measurement

Measurement probabilities:

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2$$

After measurement, the state collapses to the observed eigenstate.

1.2 Parametrized Quantum Circuits (PQC)

Parametrized Quantum Circuits (PQCs) are adjustable parameters in their gates. These parameters are typically optimized using classical algorithms in a hybrid approach. PQCs are widely used in variational quantum algorithms like VQE and QAOA. They are suitable for near-term quantum devices with limited coherence. By tuning parameters, PQCs aim to minimize a cost function and solve complex problems.

These are the foundation of:

- Variational Algorithms (VQE, QAOA)
- Quantum Machine Learning

General Form

$$|\psi(\theta)\rangle = U(\theta)|0\rangle$$

where θ represents tunable parameters.

Typical parametrized gates:

$$R_x(\theta), \quad R_y(\theta), \quad R_z(\theta)$$

Core Idea

Optimization problem:

$$\min_{\theta} \langle \psi(\theta) | H | \psi(\theta) \rangle$$

This is known as **variational optimization**.

1.3 Variational Quantum Eigensolver (VQE)

The Variational Quantum Eigensolver (VQE) is a hybrid quantum-classical algorithm used to estimate the ground state energy of a quantum system. It combines a parametrized quantum circuit with a classical optimizer to minimize the energy expectation value. The quantum computer prepares trial states, while the classical computer updates the

parameters iteratively. VQE is especially useful for problems in quantum chemistry and materials science. It is well-suited for noisy intermediate-scale quantum (NISQ)[1] devices.

Objective

To find the ground state energy of a Hamiltonian (H) of the problem.

Eigenvalue Problem

$$H|\psi\rangle = E|\psi\rangle$$

Strategy

1. Choose a parametrized ansatz
2. Compute expectation value
3. Minimize using classical optimization

$$E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle$$

Physical Interpretation

Approximation of the lowest eigenvalue of H .

1.4 Quantum Approximate Optimization Algorithm (QAOA)

The Quantum Approximate Optimization Algorithm (QAOA) is a hybrid quantum-classical algorithm designed to solve combinatorial optimization problems. It uses a parametrized quantum circuit with alternating operators to explore the solution space. A classical optimizer adjusts the parameters to maximize or minimize a given cost function. QAOA is particularly effective for problems like Max-Cut and other graph-based optimizations. It is well-suited for implementation on noisy intermediate-scale quantum (NISQ) devices.

Applications

- Combinatorial optimization (e.g., Max-Cut)

1.4.1 State Construction

$$|\gamma, \beta\rangle = e^{-i\beta H_M} e^{-i\gamma H_C} |+\rangle$$

where:

- H_C : cost Hamiltonian
- H_M : mixing Hamiltonian

Key Idea

Alternate between:

- Cost evolution
- Mixing evolution

This resembles:

- Trotterization
- Controlled quantum dynamics

1.5 Simulation Of Quantum Systems

Simulation of Quantum Systems involves using computational methods to model the behavior of quantum mechanical systems. Quantum computers can efficiently simulate systems that are intractable for classical computers. This is particularly important in quantum chemistry, materials science, and condensed matter physics. Techniques such as PQCs and algorithms like VQE are often employed for these simulations. Such simulations enable the study of molecular structures, reaction dynamics, and physical properties.

Time Evolution

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle$$

Methods

- Exact matrix exponentiation
- Trotter-Suzuki decomposition
- Variational simulation

Connection to Physics

Time evolution follows the Schrödinger equation:

$$i\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle$$

Mathematical Foundation

6.1 Linear Algebra

Complex Vector Spaces

- Orthonormal basis
- Inner product:

$$\langle\psi|\phi\rangle$$

Linear Operators

Hermitian operators:

$$A = A^\dagger$$

These represent physical observables.

Eigenvalues and Eigenvectors

$$A|v\rangle = \lambda|v\rangle$$

Applications:

- Energy levels
- Measurement outcomes

Spectral Decomposition

$$A = \sum_i \lambda_i |v_i\rangle\langle v_i|$$

6.2 Quantum Mechanics

Postulates

1. State: vector in Hilbert space
2. Observables: Hermitian operators
3. Measurement: eigenvalues
4. Evolution: unitary
5. Composite systems: tensor product

Tensor Product

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

Foundation for:

- Entanglement
- Multi-qubit systems

Measurement Operators

$$P_i = |v_i\rangle\langle v_i|$$

Final Connection

All concepts reduce to four pillars:

1. **Representation:** state vectors
2. **Transformation:** unitary operators
3. **Measurement:** Hermitian operators
4. **Optimization:** parameter tuning

Study Roadmap

Phase 1 — Mathematics

- Complex vector spaces
- Inner product
- Eigenvalues and eigenvectors
- Hermitian operators

Phase 2 — Quantum Mechanics

- Postulates
- Measurement theory
- Time evolution
- Composite systems

Phase 3 — Quantum Computing

- Qubits
- Quantum gates
- Circuits

Phase 4 — Variational Methods

- Parametrized circuits
- VQE
- QAOA

Phase 5 — Implementation

- Python-based quantum frameworks
- Simulation and experimentation

Referências

1. Bharti, K. *et al.* Noisy intermediate-scale quantum algorithms. *Rev. Mod. Phys.* **94**, 015004. <https://link.aps.org/doi/10.1103/RevModPhys.94.015004> (1 2022).